

KUSHIRO KENEBETSU, Japan

WMO: 474180

Lat: 42.9861N

Lon: 144.3772E

Elev: 40

StdP: 100.85

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-13.1	-19.3	0.7	-12.2	-17.6	0.8	-10.6	16.2	-1.4	14.4	-0.6	3.6	20	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	23.9	20.4	22.6	19.8	21.5	19.3	21.5	22.9	20.6	22.0	19.8	21.1	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	15.8	22.2	20.1	14.8	21.4	19.2	14.1	20.8	63.1	23.0	59.9	22.2	57.0	21.1	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 14.4	12.3	10.6		-18.4	27.4	2.3	1.8	-20.0	28.7	-21.4	29.8	-22.7	30.8	-24.4	32.1	
(5)			DB	-18.7	22.5	2.3	1.0	-20.4	23.3	-21.7	23.9	-23.0	24.5	-24.7	25.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-5.2	-5.0	-0.6	4.1	8.7	12.4	16.5	18.6	16.4	10.6	4.3	-2.2
	DBStd	8.71	3.21	3.42	3.02	2.59	2.80	2.64	2.49	2.46	2.72	3.06	3.55	3.42
	HDD10.0	2044	472	419	329	177	58	7	0	0	0	30	174	378
	HDD18.3	4334	730	652	588	427	298	178	67	28	69	239	422	636
	CDD10.0	810	0	0	0	1	18	80	202	266	193	49	2	0
	CDD18.3	59	0	0	0	0	0	1	11	35	13	0	0	0
	CDH23.3	69	0	0	0	0	0	2	11	44	12	0	0	0
	CDH26.7	4	0	0	0	0	0	1	0	3	1	0	0	0
(14) Wind	WSAvg	4.6	4.9	5.0	5.3	4.9	4.6	3.8	3.4	3.7	4.2	5.0	5.5	5.3
(15) Precipitation	PrecAvg	1109	43	28	52	85	120	110	124	140	164	107	71	64
	PrecMax	1569	133	78	113	206	254	287	284	432	279	246	172	147
	PrecMin	822	4	4	8	12	23	28	20	33	66	25	8	12
	PrecStd	190	30	22	29	51	55	72	63	84	58	59	41	36
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	4.4	8.5	13.9	19.4	21.7	24.7	26.7	24.9	19.7	14.3	9.4
		MCWB	2.9	1.8	4.3	7.5	12.5	17.1	20.3	21.8	21.1	15.1	11.6	7.7
	2%	DB	2.7	2.9	6.1	11.3	16.3	19.2	22.8	24.4	22.7	18.0	13.3	7.3
		MCWB	1.1	1.0	3.1	6.7	11.1	15.6	19.3	20.9	19.7	14.7	11.1	5.5
	5%	DB	1.6	1.7	4.7	9.5	14.3	17.4	21.3	23.0	21.6	16.9	12.2	5.4
		MCWB	-0.2	-0.1	2.2	6.0	10.3	14.6	18.8	20.3	19.2	14.2	10.0	3.0
	10%	DB	0.6	0.7	3.6	8.1	12.8	16.1	20.1	22.0	20.6	15.8	11.0	3.7
		MCWB	-1.4	-0.9	1.5	5.5	9.9	14.0	18.4	20.1	18.3	13.2	8.7	1.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	2.9	5.5	9.2	13.9	18.0	21.2	22.7	22.5	17.3	13.1	8.2
		MCDB	4.1	3.6	7.4	11.6	17.1	20.5	23.1	24.5	23.5	18.3	13.8	8.9
	2%	WB	1.6	1.2	3.9	7.7	12.3	16.3	20.1	21.8	21.1	15.9	11.8	5.8
		MCDB	2.5	2.3	5.4	9.9	14.7	18.3	21.7	23.2	22.2	17.1	12.8	7.1
	5%	WB	0.2	0.3	2.7	6.8	11.2	15.2	19.4	21.1	19.9	14.9	10.4	3.5
		MCDB	1.3	1.5	4.1	8.7	13.2	16.6	20.8	22.5	21.0	16.2	11.8	5.0
	10%	WB	-1.3	-0.7	1.7	5.9	10.3	14.3	18.5	20.3	18.7	13.8	8.9	1.5
		MCDB	0.5	0.7	3.2	7.8	12.2	15.6	19.9	21.7	20.1	15.4	10.6	3.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	8.8	6.8	6.5	6.5	5.5	5.1	5.0	6.3	8.2	8.3	8.5
		MCDBR	8.9	7.9	7.1	8.7	9.4	7.7	7.2	6.5	6.8	7.9	9.1	9.7
	5% WB	MCWBR	8.0	6.7	5.4	5.4	5.2	4.7	4.4	3.8	4.3	6.0	8.3	8.9
		MCWBR	9.1	7.7	6.8	7.8	7.7	6.6	6.3	5.7	5.5	6.9	8.9	9.8
(40) Clear-Sky Solar Irradiance	taub		0.322	0.368	0.446	0.505	0.522	0.496	0.483	0.466	0.412	0.394	0.368	0.334
		taud	2.186	2.037	1.919	1.848	1.891	2.053	2.159	2.224	2.330	2.245	2.224	2.204
	Ebn at Noon		787	800	779	765	766	788	794	795	814	772	726	731
		Edh at Noon	95	132	169	196	193	165	147	133	111	107	93	87
(44) All-Sky Solar Radiation	RadAvg		1.99	2.91	3.92	4.63	4.89	4.82	4.38	4.06	3.58	2.90	1.97	1.61
	RadStd		0.14	0.19	0.27	0.34	0.42	0.43	0.40	0.36	0.31	0.19	0.14	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.42	N/A	N/A	N/A	N/A	+0.78	N/A	-117	+86	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-66	N/A	N/A

Nomenclature: See separate page